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Q-4 What is Generics? How to Create a generic class & methods. Explain with it a java Example?

Generics means Parameterized types. The idea is to allow type (integer, String ... etc, & user-defined types) to be a parameter to methods, classes & interfaces. Using generics, it is possible to create classes that work with different data types.

Ans 5

JDBC Architecture

- JDBC Consists of two parts —

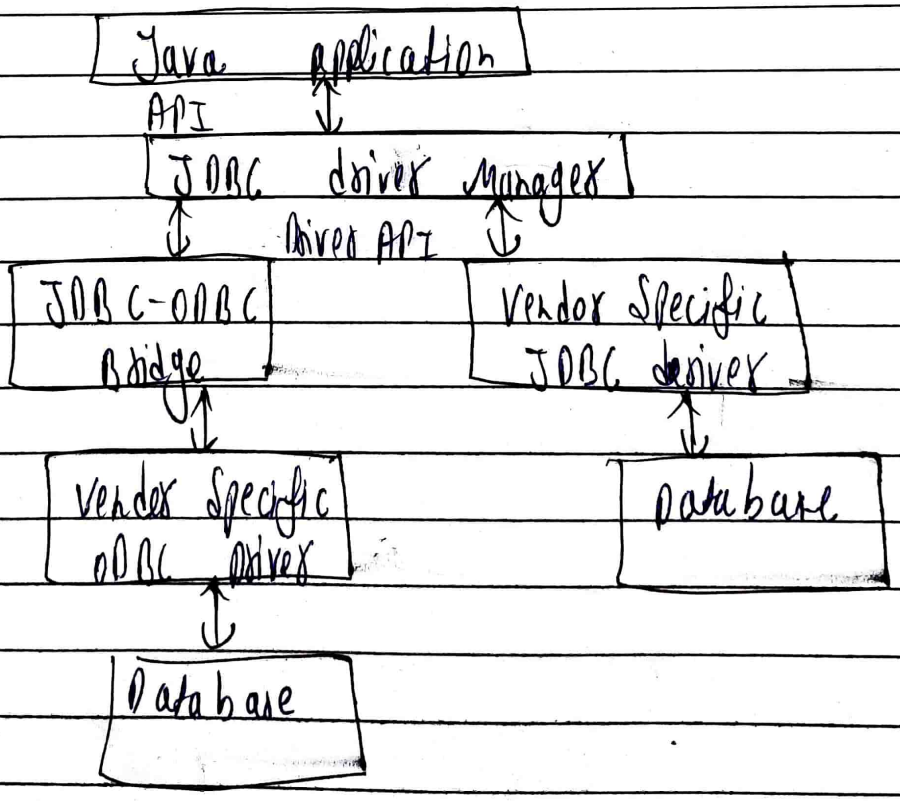
→ JDBC API a purely java based API

→ JDBC Driver Manager, which communicates with vendor-specific drivers that perform the real communication with the database

- Translation to the vendor format to occurs on the client —

→ No changes needed to the server

→ Driver (translation) needed on client.



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Ans → 6 ① The throw → ① The throw keyword is use inside a function. It's use when it is required to throw an Exception logically.

② The Throw keyword is used to throw an exception explicity it can throw only one exception at a time

② The throws → ① keyword is used in the function Signature it is used when the function has some statement that can led to exceptions.

② keyword can be used to declare multiple exception seprated by a comma whichever exception occurs, if matched with the declared ones, is thrown automatically then.

Ex → ① class test {
public static void main(String [] args)
{
System.out.println(1/0); } }

② class test
{ public static void main (String [] args)
{
Thread.sleep (1000);
System.out.println ("Hello");
} }

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Ques 7 What are the J.D.B.C API Components?

Ans → JDBC Stands for Java database Connectivity. It established Connection to access the database.

① Driver Manager → The JDBC drivers & established the Connection to the database. It uses the jdbc URL to established the Connection & data base. It returns the Connection object on a Successfull data base Connection

② Connection → The Connection object represents the Session to the Connected database. The Create Statement object to send SQL Statements to the database

③ Statement → The Statement object is used for executing the SQL Statements & returns the SQL query results.
Eg: Stmt. executeQuery(SqlQuery)

④ Result Set → The Result Set object refers to the SQL query result table. The Result Set object maintains a Cursor pointing to the Current row of the query result table. We Can use a while loop to iterate through the Result Set.

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Ans:- The execute() method: This method is used to execute SQL DDL Statements, it returns a boolean value specifying weather the ResultSet object (an be retrieved.

Execute query(): This method is used to execute Statements that returns tabular data (example Select) It returns an object of the class resultSet.

execute update(): This method is used to execute Statements such as insert, update, delete It ~~update~~ returns as integer value representing the number of rows affected.

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Q) What is thread Scheduler. On which Algo it's works?

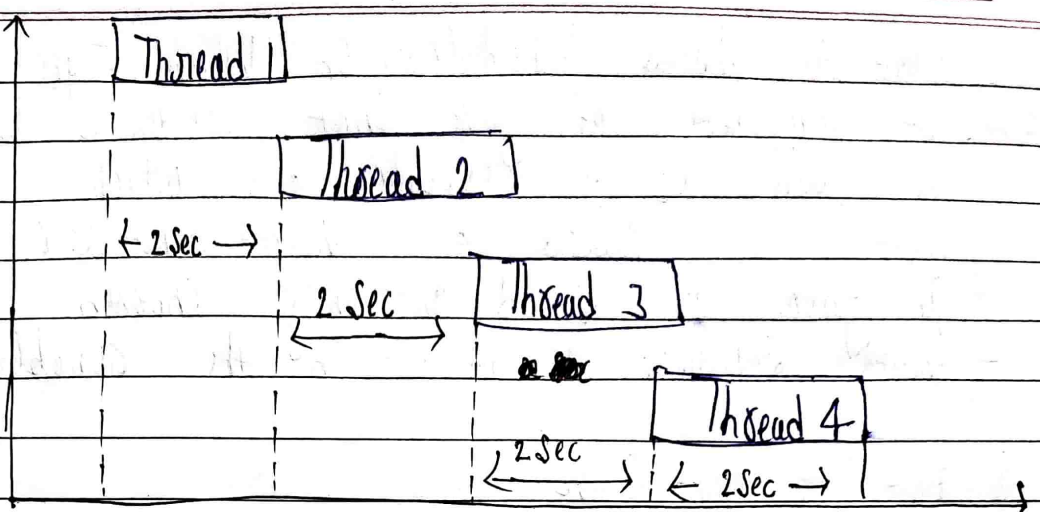
Ans) A Component of java that decides which thread to run or execute and which thread to wait is called a thread Scheduler in java. In java a thread is only choosen by a thread Scheduler if it is in the runable state.

① First Come first Serve
② Time-Slicing Scheduling
③ Preemptive-priority Scheduling

① First Come first Serve → In the Scheduling algorithm, the Scheduler picks the threads that arrive first in the runable queue.

Threads	time of arrival
t_1	0
t_2	1
t_3	2
t_4	3

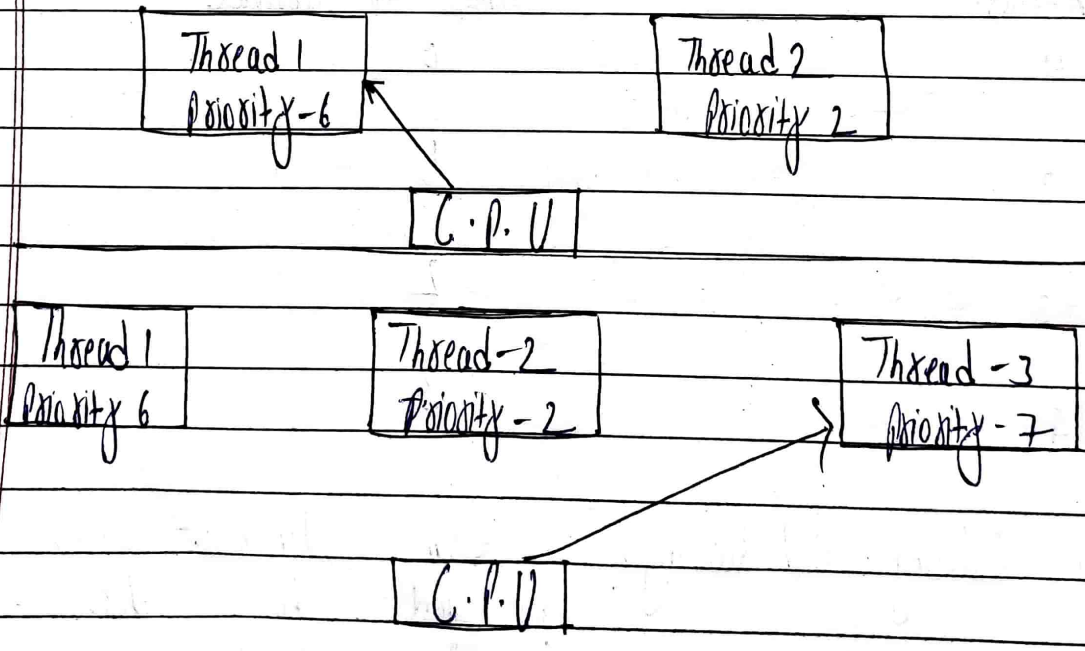
② Time Slicing Scheduling → Some times - Slices are proceeded to the threads So that after some time



Time Slicing Scheduling

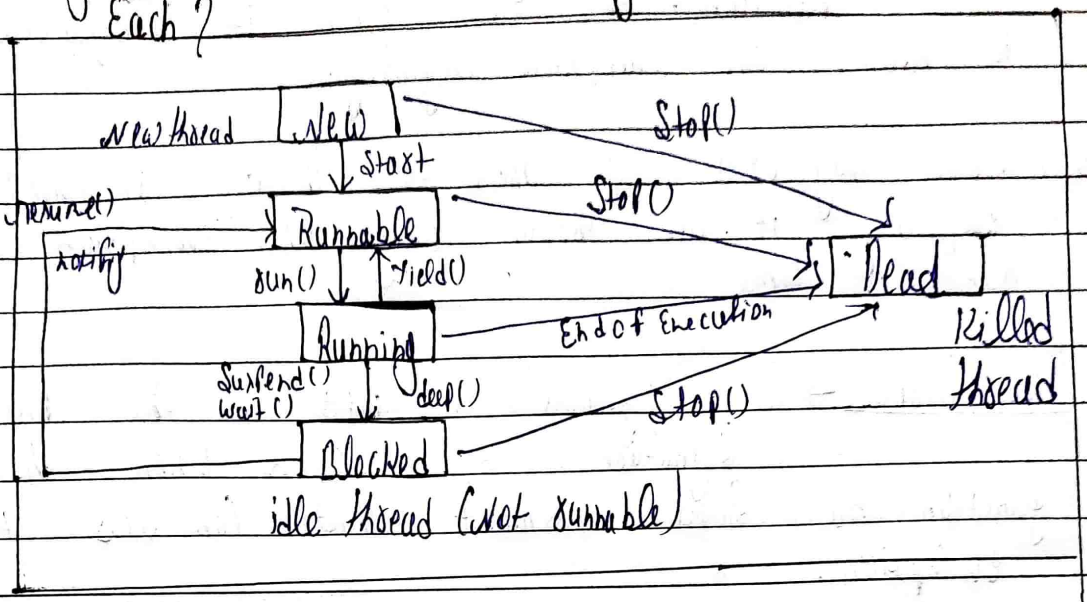
③ Preemptive - priority Scheduling →

The name of the Scheduling algorithm denotes that the algorithm is related to the priority of the threads.



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Q→ Lifecycle of Thread ? How many States are there Explain Each ?



① New → When we create a thread object using Thread class, thread is born and is known to be in Newborn State.

② Runnable State → Runnable State means a thread is ready for execution. When the start() method is called on a new thread, thread enters into a runable State.

③ Running State → Running means processor (CPU) has allocated time slot to thread for its execution.

④ Blocked State → A thread is considered to be in the blocked State when it is Suspend sleeping waiting Some time in order Satisfy Some condition

⑤ Dead → A thread dies or moves into dead State automatically when its run method complete the execution of Statement.